

MA3K7 Big mini project rubric

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1 Entry

I WANT	- I would like to investigate whether there exists a winning strategy to the "pub game" for Ali and if so, what it is - Unfortunately, I cannot recall any previous problem or area of mathematics that I am able to relate this problem to. The only problem that I have seen relating to a circular structure as such is the Josephus problem, but the setup for that is rather different
NOTATION	Let us call Ali and Beth: A and B respectively. We also say that a game of size n has n glasses on the table
ASSUMPTION	Let us assume that both A and B are acting rationally and intelligently
I KNOW	- A is able to distribute 0.5 units across any glasses on his move, B is able to empty any two adjacent glasses on her move - It is clear after some thought, that it is impossible for A to win if $n < 3$, since in this case B can simply empty all available glasses on her move
STUCK	Otherwise, it is not clear what the best strategy is for either player, since there are infinitely many moves for player A
I KNOW	Occam's razor tells us that we should consider the simplest ways of filling the cups and not worry about obscure and irrational distributions of water
SPECIALISATION	So, let us first investigate the case $n = 3$

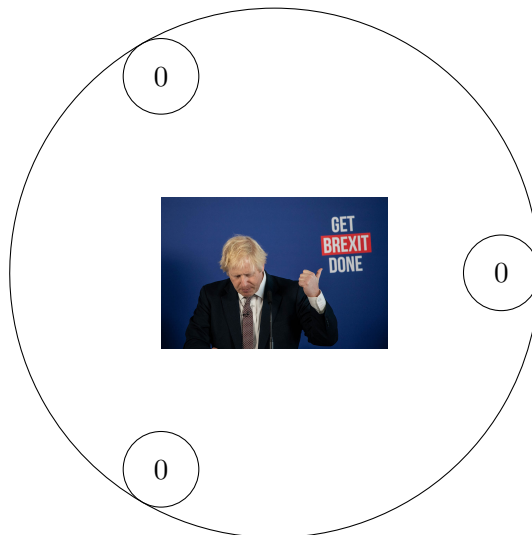


Figure 1: Birdseye view of setup for the game with $n = 3$, with an enthralled audience onlooking

INTRODUCE	It will be inconvenient to reproduce the above diagram for many moves, so we simplify it as $(0,0,0)$, where the volume in the i^{th} cup is represented by the i^{th} component of the vector
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- We first observe that if A only distributes the water among one or two glasses, they will be immediately removed by B
- Hence, A must not leave any of the 3 cups empty on his starting move
- I also notice that if A doesn't distribute symmetrically among all 3 cups, then B will be able to empty the 2 larger cups, leading to less water remaining for A than if he did otherwise
- Hence A should start by pouring the $\frac{1}{2}$ unit equally into the 3 cups, I.e. $\frac{1}{6}$ units into each
- Game state is now: $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6})$
- Then the game will move to state $(\frac{1}{6}, 0, 0)$, regardless of B's choice (WLOG)
- By a similar reasoning to the last move, we first aim to bring all cups up to the same volume, which leaves: $\frac{1}{2} - 2 * \frac{1}{6} = \frac{1}{6}$ volume to distribute. Again we look to split equally to maximise the remaining cup after B's move, I.e. $\frac{1}{18}$ vol amongst glasses
- This leaves us in game state $(\frac{2}{9}, \frac{2}{9}, \frac{2}{9})$
- Which again will have to go to $(\frac{2}{9}, 0, 0)$ after B's move

- This move structure seems to have an element of iteration, perhaps we could exploit this to set up a recurrence relation

- If we keep applying the same strategy, we can translate it to the following equation: $X_{n+1} = X_n + (\frac{1}{2} - 2X_n) \cdot \frac{1}{3}$, simplifying gives: $X_{n+1} = \frac{1}{3}X_n + \frac{1}{6}$ with $X_0 = 0$
- Let us clarify the origin of the terms in the above equation:
 - Here X_n is the non-zero component of the state vector after B's n^{th} move
 - $(\frac{1}{2} - 2X_n)$ represents the volume left after in A's small cup after he equalises the three pint glasses
 - The factor of $\frac{1}{3}$ is to signify that we split the excess uniformly between all cups
- Let us now solve this recurrence relation to see if we can deduce anything about the endgame state:

I know from previous study that the general solution of this first order difference equation is given by:

$$X_n = A \cdot \left(\frac{1}{3}\right)^{(n)} + B$$

Where A and B are constants to be found using our boundary conditions $X_0 = 0$ and $X_1 = \frac{1}{6}$. Let us continue with this:

$$\begin{aligned}
 0 &= A + B && \text{(from condition } X_0 = 0) \\
 \frac{1}{6} &= \frac{A}{3} + B && \text{(from condition } X_1 = \frac{1}{6}) \\
 \implies A &= -\frac{1}{4}, B = \frac{1}{4} && \text{(solutions to the above simultaneous equations)}
 \end{aligned}$$

So our final closed form solution to X_n is given by:

$$X_n = \frac{1}{4} - \frac{3^{-n}}{4}$$

- We note crucially that $\lim_{n \rightarrow \infty} X_n = \frac{1}{4}$
- Recall that X_n represents the volume of the fullest glass after B's n^{th} move, assuming A plays his **best** strategy
- This means that A will never be able to fill a glass more than $\frac{1}{4}$ full, so can never win when $n = 3$:(

2 Attack

- Before we begin further specialisation to deduce more patterns, I would like to make some comments about the result we found in the entry phase(A cannot win if $n = 3$):
 - As n increases, the proportion of the total water volume that B can remove in any given move will decrease
 - This means that the case $n = 3$ should be the lower bound(non-trivial) on the maximum volume A can achieve in any single cup
 - We also note that in the case $n = 3$, we did not need to utilise the restriction that B can only remove **adjacent** cups
 - Now we should be able to exploit this property to find a strategy for A that is more effective than the symmetric distribution method

Let us analyse the case $n = 4$, with a similar strategy as previously to begin with

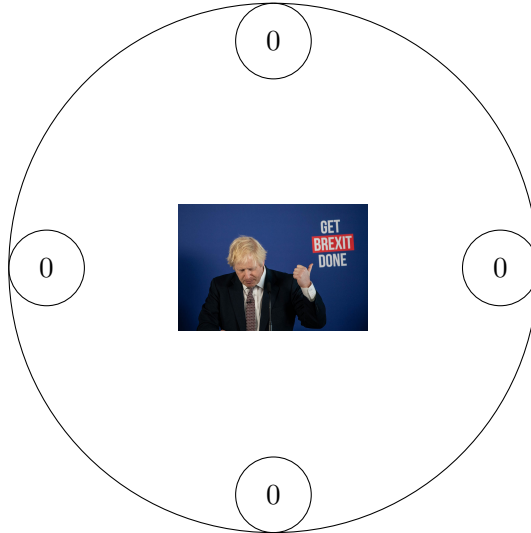


Figure 2: Initial setup for the game with $n = 4$, with a rather excited spectator...

- Since we are looking to maximise the least filled cup in this strategy, A begins the game with: $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8})$
- WLOG this leads to the two following states after B's move : $(\frac{1}{8}, \frac{1}{8}, 0, 0)$ and $(\frac{1}{8}, 0, \frac{1}{8}, 0)$
- But wait, of course this will not work :(. We are trying the same strategy but splitting that $\frac{1}{2}$ cup into 4 parts rather than 3, which will be even slower to build up volume
- Instead, let us attempt to exploit B's adjacent selection restriction to create a more potent strategy

AHA

- So we notice that WLOG, we can effectively split the table into two (pairs) and B can only ever remove water from either half
- Bearing this in mind, let A begin as follows: $(\frac{1}{4}, 0, \frac{1}{4}, 0)$
- Then WLOG, B's move takes us to: $(\frac{1}{4}, 0, 0, 0)$
- Notice how B can only ever remove one of two filled cups, never both

TRY

- So we aim to equally maximise these two cups that are separated by one
- A must first fill cup 3 (c_3) to equal c_1 , this leaves over $\frac{1}{4}$ to be split equally between them
- So our next move from A gives us: $(\frac{3}{8}, 0, \frac{3}{8}, 0)$
- Like the case $n = 3$, it is clear this will produce a recurrence relation, so let us express and solve it:

We have $X_{n+1} - X_n + X_{n+1} = \frac{1}{2}$, simplifying which gives: $X_{n+1} = \frac{1}{2}X_n + \frac{1}{4}$ with $X_0 = 0$

Like last time, we denote the volume of the remaining filled cup after B's n^{th} move by X_n . The general solution of this first order difference equation is given by:

$$X_n = A \cdot \left(\frac{1}{2}\right)^n + B$$

Where A and B are constants to be found using our boundary conditions $X_0 = 0$ and $X_1 = \frac{1}{4}$. Let us continue with this:

$$\begin{aligned} 0 &= A + B && \text{(from condition } X_0 = 0) \\ \frac{1}{4} &= \frac{A}{2} + B && \text{(from condition } X_1 = \frac{1}{4}) \\ \implies A &= -\frac{1}{2}, B = \frac{1}{2} && \text{(solutions to the above simultaneous equations)} \end{aligned}$$

So our final closed form solution to X_n is given by:

$$X_n = \frac{1}{2} - \frac{2^{-n}}{2}$$

- Realising that $\lim_{n \rightarrow \infty} X_n = \frac{1}{2}$, tells us that A will never be able to fill a glass more than $\frac{1}{2}$ full, so can never win with this strategy
- Although this method of maximising alternating cups has not worked with $n = 4$, it may work for larger games. This is because B will have more options of which pair to eliminate, which will leave more pairs untouched on any move
- However, this strategy does seem optimal to me, so I conjecture as follows:

MAYBE

Conjecture

It is impossible for A to win when $n = 4$

INTRODUCE

- Before we begin our justification, let us reintroduce some notation

- Namely, we label the cups c_i , corresponding to the i^{th} cup on the table

JUSTIFICATION

- Note that any move by B will remove water from c_k and c_{k+1} , for some $k \in \{1, 2, 3, 4\}$
- Since we are looking to optimise the volume in a single given cup, A must maximise either c_k or c_{k+1} , which is homologous to only filling alternating cups

AHA

- This then effectively reduces the game to one played with two cups, where B is only able to empty one of them

TRY

- We can translate this to a more graphical/numerical problem to provide a different perspective
- Let us instead view this as a game where A is trying to escape the box $([0, 1] \times [0, 1])$ when restricted to the space $(x, y) \in \mathbb{R}^2 : x > 0, y > 0$, while starting at the $(0, 0)$
- On A's move, he can move a total positive displacement of $1/2$ in the L^1 norm
- On B's move, she can reduce either co-ordinate of A's position to 0. This is geometrically equivalent to projecting A onto either axis

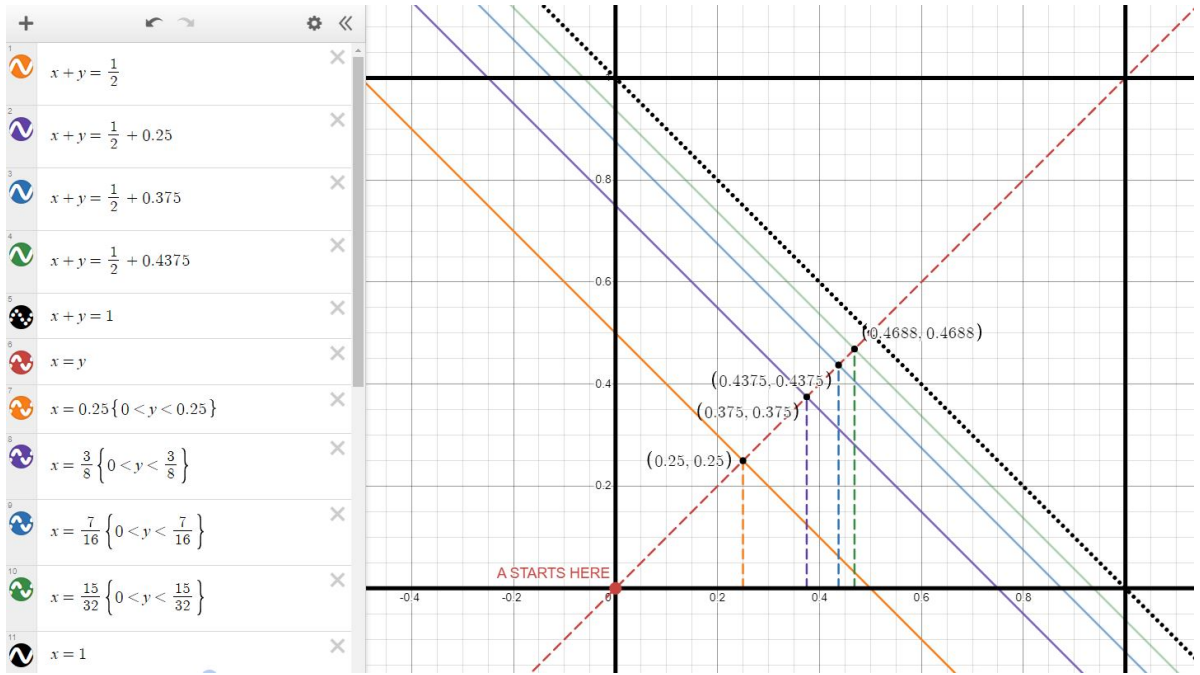


Figure 3: visualisation of the transformed problem using desmos

- Our proof will follow a greedy algorithmic approach
- On A's first move, he can achieve an optimal distance by moving to the point $(\frac{1}{4}, \frac{1}{4})$
- The possible moves can be encapsulated by the line $X + Y = \frac{1}{2}$ (orange line), which reflects the restriction of $\frac{1}{2}$ displacement in the L^1 norm
- This is because any other move would produce a larger and a smaller distance along either axis, which would be exploited by B when projecting to result in a final point who's distance is less than $\frac{1}{4}$
- After B's projection we are at $(\frac{1}{4}, 0)$, WLOG from symmetry, A's possible moves are now encapsulated by the line $X + Y = \frac{1}{2} + \frac{1}{4}$ (purple line). By a similar argument as previously, we aim for the symmetric point, which is $(\frac{3}{8}, \frac{3}{8})$ in this case

- Now we repeat this process but starting from the point $(\frac{3}{8}, 0)$...
- This forms a difference equation as follows: $2X_{n+1} = \frac{1}{2} + X_n$, $X_0 = 0$. Here X_n is the non-zero co-ordinate after B's projection
- We notice this is the same as our difference equation from previously, which has property: $\lim_{n \rightarrow \infty} X_n = \frac{1}{2}$
- So clearly A has no way of escaping the box, and does indeed approach the limit as can be seen in the graph (black dotted line)
- Since this problem is equivalent to the pub game with $n = 4$, we conclude that our conjecture is indeed correct $\square$

AHA

REFLECT

Just to recap what we have show so far:

- it is impossible for A to win games with $n \leq 4$
- For all **even** sized games, any winning strategy must incorporate alternately filling cups (but may improve upon this)
- We were able to reduce the problem dimensions and thus transform it to a more geometric approach
- We also noticed that the alternating cup strategy may result in a win with larger n

KEY STEP

Bearing this in mind, we look to attack the case $n = 6$:

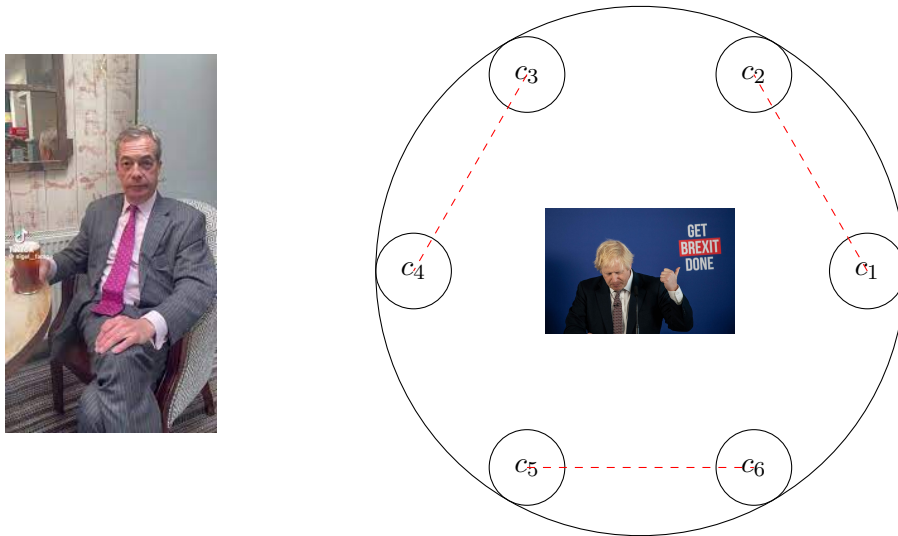


Figure 4: Our game with $n = 6$, the dashed lines represent paired cups. Our alternating strategy looks to only ever fill one cup in each pair. Our crowd is now truly on the edge of their seats, as can be seen...

- First we reduce the dimensions as with $n = 4$, i.e we realise this game with 3 cups and B emptying any one cup on her move
- recall, we can do this due to the alternating cup strategy we employ (and proved is necessary earlier)
- So A's first move should be to equalise all cups, giving $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6})$

- WLOG this leads to $(\frac{1}{6}, \frac{1}{6}, 0)$ after B's first move

- As with the previous games , we now look to level the cups. But this time, let us redistribute the excess into only two of the cups rather than 3. This is better than splitting equally and is not as risky as maximising just one

- So A's next move leads to: $(\frac{2}{6}, \frac{2}{6}, \frac{1}{6})$, which goes to $(\frac{2}{6}, 0, \frac{1}{6})$ after B's move (WLOG)

- Notice that if we now equalise all cups, we have reached an exact double of the state after move 1

- So lets do so: $(\frac{2}{6}, \frac{2}{6}, \frac{2}{6})$, which leads to $(\frac{2}{6}, \frac{2}{6}, 0)$ after B's move (WLOG)

- This is a critical point, since we are now able to force two cups over the $\frac{1}{2}$ mark ; one of which we can overfill !!!

- A's next move is add $\frac{1}{4}$ cup to the two, giving $(\frac{7}{12}, \frac{7}{12}, 0)$. Surprise Surprise, this goes to $(\frac{7}{12}, 0, 0)$ after Bs move (WLOG)

As A carefully pours the last half of the glass into the pint, a crescendo of anticipation fills the room. With each drop, the tension mounts until, with a sudden surge, the liquid cascades over the brim, a poignant reminder of A's relentless pursuit to fill the glass, a pursuit that seems to have lasted an eternity.

Nigel, who had been observing with a wry smile, couldn't resist chiming in, "Well, would you look at that! You've finally achieved the impossible, mate. It's like watching Brexit negotiations ; a lot of drama, but you got there in the end! Cheers to your perseverance!"



- Well, to be exact the final state is: $(\frac{13}{12}, 0, 0)$, or $(\frac{13}{12}, 0, 0, 0, 0, 0)$ in full form

- Let us gather our thoughts and recap where we have reached with the problem:

– We have found that A cannot win for $n \leq 4$, but can win for $n = 6$

– It is now evident to me that A can actually win for $n \geq 6$, since we can simply ignore the extra cups and play the winning strategy we employed in game $n = 6$

– Therefore, the only case left to analyse is that of $n = 5$

– I cannot immediately see any winning strategy , but we will get stuck in and see what we can deduce...

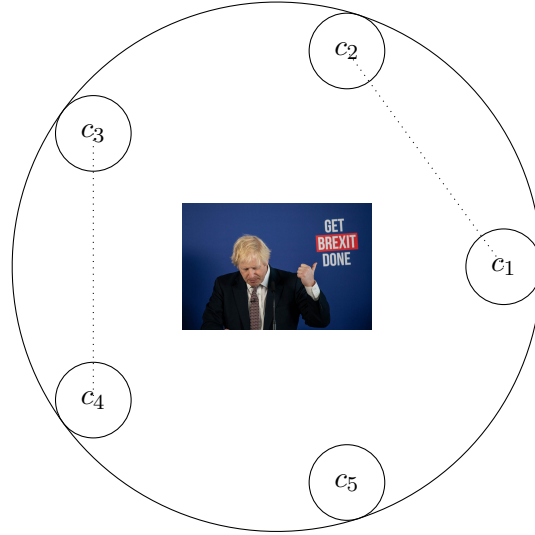
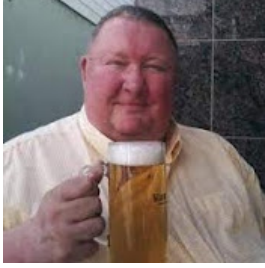


Figure 5: Birdseye view of game $n = 5$. With a pint in hand, Barry, fueled by banter and brews, watches A's every move, ready to tease at the first sign of failure...

Conjecture

It is impossible for A to win when $n = 5$

AHA

- Note that on any given starting move by A, there will be 2 adjacent cups being filled

I KNOW

- This is due to the pigeonhole principle I have come across in previous problems
- Because of this, A's optimal first move only utilises c_1 and c_3 (WLOG)

JUSTIFYING

- This has effectively reduced the game to that being played with $n = 4$, since the extra cup will always be adjacent to one of the "other" 4
- So even if B plays optimally while ignoring the 5th cup, she will end up emptying it free of charge
- Therefore A's optimal strategy is to ignore the fifth cup, thereby playing the game $n = 4$
- However, we previously showed that A **cannot** win $n = 4$
- So our conjecture is correct, A cannot win $n = 5$!

3 Review

CHECK

- In conclusion, we found that Ali is able to force a win for all $n \geq 6$

EXTENDING

- One way that we can look to extend this problem is to consider giving an advantage to Beth
- Let us imagine now that Beth can remove b adjacent cups, instead of just 2
- Notice that we can apply a similar "alternating" strategy as we proved in our attack, to show that A should ignore any cups between c_1 and c_{b+k} for all $k \in \mathbb{Z}$

AHA

- This then reduces the dimension of the game to one being played with $\lfloor \frac{n}{b} \rfloor$ cups and Beth can remove 1 cup on her turn

KEY STEP

- This is exactly the underlying transformation we have used to compare different games in an identical set up
- As we proved in the attack, this transformed game can be won with 3 or more cups. Equivalently, the original game can be won by Ali whenever $n \geq 3 \cdot b$
- You'll notice that throughout the attack phase, I was mainly focusing on deducing A's optimal move since B's optimal move was evident

EXTENDING

- So i decided to utilise python to simulate games where B played randomly, using Monte-Carlo style simulation

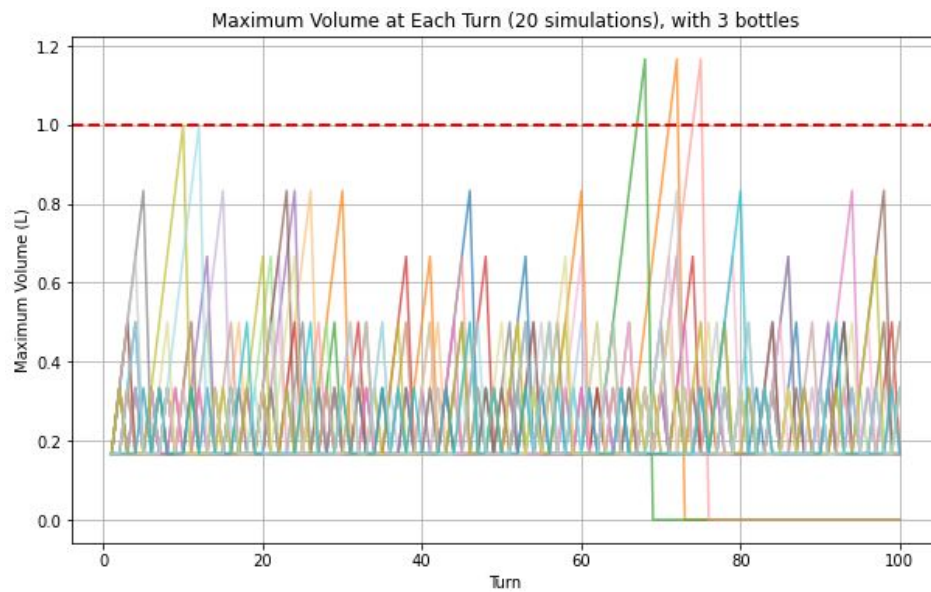


Figure 6: A's strategy: split water equally between cups, B's strategy: random

- Although we proved it impossible for A to win with $n = 3$, it is clear from the above graph that A can indeed win if B does not play intelligently
- Interesting however, it is still exceedingly unlikely for A to win despite B's randomness. A only goes only wins 3 times, after 100 turns in 20 games, which suggests a $\frac{3}{2000}$ probability given A's equal splitting (not optimal)

EXTENDING

- To take this idea even further, I expanded the code to simulate this game with different sized games ($2 \leq n \leq 10$)
- By counting the number of games won up to any given turn, I was able to produce the insightful plot below
- As expected, larger sized games were quicker to convert to wins (water over-spilling)

WHY

- This is because B's "error" from the *best* move is amplified as the number of glasses increases
- Curiously, the plots seem to approach a limiting curve as the number of pint glasses increases. This curve demonstrates logarithmic growth properties with respect to the number of turns

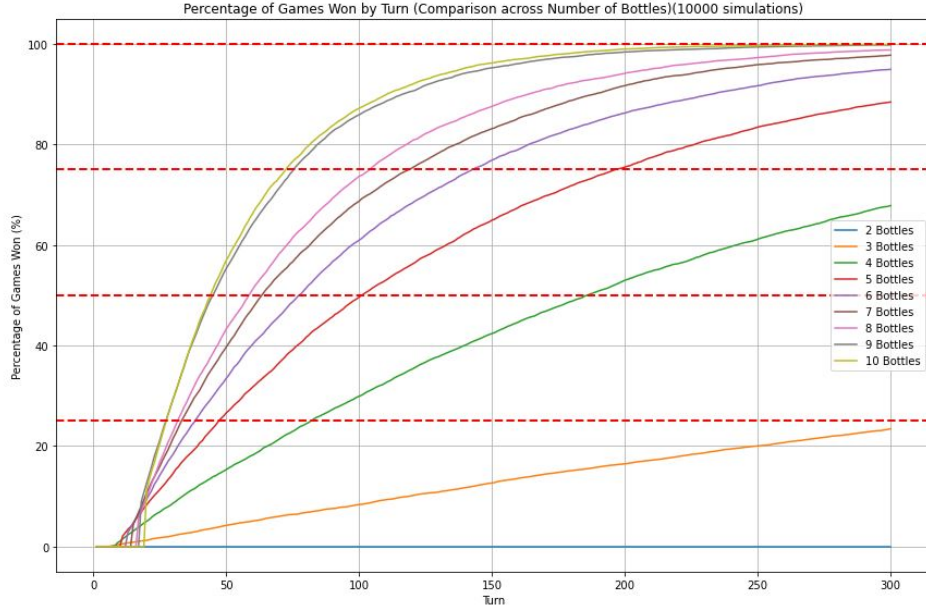


Figure 7: graph showing how quick different sized games are to win, given B acts randomly

- There is another way that we can extend this problem, that is to alter the initial conditions and restrictions provided
- One way this can be done is to change the size of the small cup A uses to distribute water
- Notice that we can simply generalise the difference equations we produced in our attack phase to this end:

$$\begin{aligned}
 X_{n+1} &= X_n + (V - X_n \cdot (N - 1)) \cdot \frac{1}{N} \\
 &= X_n - X_n \cdot \frac{N - 1}{N} + \frac{V}{N} \\
 &= \frac{V}{N} + \frac{1}{N} \cdot X_n
 \end{aligned}
 \quad , \text{ with } X_0 = 0$$

- Here V is the volume in A's pouring cup, which is originally $\frac{1}{2}$. N is the number of pint glasses in the game
- We can use this equation to plot game states given A's optimal moves under an equalising strategy without alternating (same as we used in $n = 3$)
- Aside from the beautiful colours, we notice that there seems to be relationship between the number of pint glasses and the "levelling volume", namely that the levelling volume is exactly one less than the number of glasses
- By levelling volume, we mean the required size of small glass such that A is able to fill a pint glass to brim if he plays the equal splitting strategy

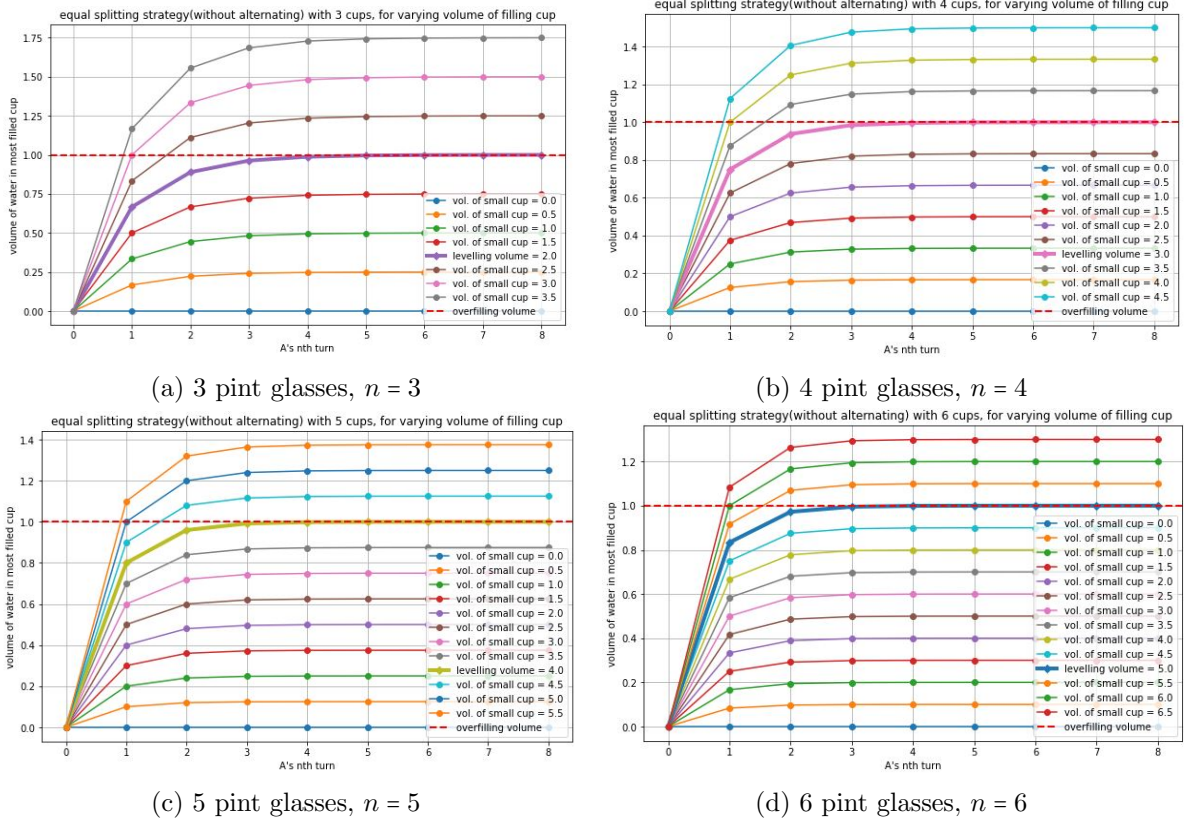


Figure 8: rate of filling for games size $3 \leq n \leq 6$, with varying volumes A can distribute

- This could stem from solving the master difference equation for this strategy, as described above:

The general solution of this first order difference equation is given by:

$$X_n = A \cdot \left(\frac{1}{N}\right)^n + B$$

Where A and B are constants to be found using our boundary conditions $X_0 = 0$ and $X_1 = \frac{V}{N}$. Let us continue with this:

$$\begin{aligned} 0 &= A + B && \text{(from condition } X_0 = 0) \\ \frac{V}{N} &= \frac{A}{N} + B && \text{(from condition } X_1 = \frac{V}{N}) \\ \implies A &= \frac{V}{1-N}, B = \frac{V}{N-1} && \text{(solutions to the above simultaneous equations)} \end{aligned}$$

So our final closed form solution to X_n is given by:

$$X_n = \frac{V}{1-N} \cdot \left(\frac{1}{N}\right)^n + \frac{V}{N-1}$$

We notice that if we let $V = N - 1$ then the solution simplifies to:

$$X_n = 1 - \left(\frac{1}{N}\right)^n$$

- Since the limit of this solution is 1, we can deduce that for $V > N - 1$ A will eventually win (although it may take a while depending on how much larger it is)

REFLECT	• For me, the key step in this rubric was spotting the alternating strategy. This was crucial to both simplifying larger cases by a transformation, and also aiding proofs that we otherwise confusing ($n = 5$) • Another vital realisation was that once we manage to win a game, all larger games could instantly be won. This significantly reduced the workload for me, changing a daunting problem with infinite possibilities to one of a few small tangible cases
REFLECT	• Something that separated this problem from previous ones on this course was the sheer depth of it • This meant that I had to apply many cycles of the "entry-attack-review" framework to really dig into the heart of the problem
KEY STEP	• I found that my own reflections at the end of these mini-cycles was necessary to gather my thoughts and guide the next direction of attack • They were extremely useful to clarify how far I had reached in the problem and to evaluate if any of my current approaches gave me any insights I could carry forward
PYTHON	• It was a shame that I wasn't able to utilise python to any useful end in the attack phase
BUT WHY	• Although this to me was due to the setup of the problem, given that A had uncountable infinite number of moves available to him. This is a serious problem if trying to discretize the problem into code, as well as ignoring the high likelihood of infinite loops if the game didn't terminate • However, B's moves were much easier to implement since she only had a small and finite number of possible moves • This was reflected in my simulations, where B is able to act randomly, but A must always act in a predefined manner
POSSIBLE	• I had an idea of creating a machine learning model called <i>reinforcement learning</i> to help solve this problem. The RL model would play against itself a large number of times to try and learn an optimal strategy for A(and B); you may have seen something similar with robots learning to play Pac-Man! • However implementing this was beyond my current knowledge of python :(
ASSUMPTIONS	• Unlike previous assignments, there not any assumptions I needed to make in this problem as it was well defined
REFLECT	• There were many techniques that I developed in this rubric(and module) that I will look to apply in future problems I may come across • I'd like to end by saying cheers to Mark and Siri for what was a well organised and engaging module, truly the highlight of this term. Thanks for making maths feel less like a chore and more like a thrilling adventure!

Code

The code for this assignment can be found on my GitHub page: <https://github.com/abdullah12353/MA3K7/blob/main/MA3K7%20%20bigminipro%20u2104970.ipynb>